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Purpose: Build an automated, reliable playing-card machine that can shuffle and pass cards to players

Constraints/Specs:

- i) Mechanical assembly, routing infrastructure, and electronics must be housed cleanly inside an enclosure box.
- ii) Machine operates on a single battery pack (voltage of choice)
- iii) Must be portable and weigh no more than 1.4 Kg (3 lbs)
- iv) Must have a user interface for menu navigation
- v) Needs functions to perform the following:
 - a) Shuffling
 - b) Dispense a specified # of cards to all players
 - c) Dispense a card to a player based on user input
 - d) Stop all mentioned functions via button
- vi) Any electronic hardware can be used
- vii) Should incorporate thermal cooling/venting in enclosure design
- viii) Microcontroller USB port should be accessible from outside the box
- ix) Power switch must be accessible and labeled
- x) prevent system from running on too low voltage

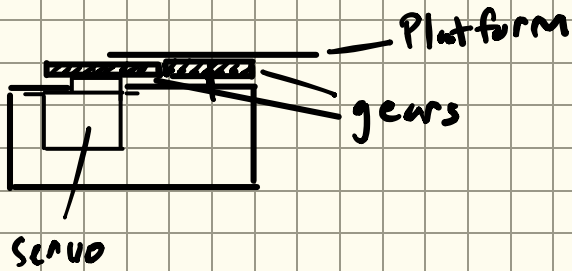
Designer: Ethan Luong

Date: 7/11/26

Witness: _____

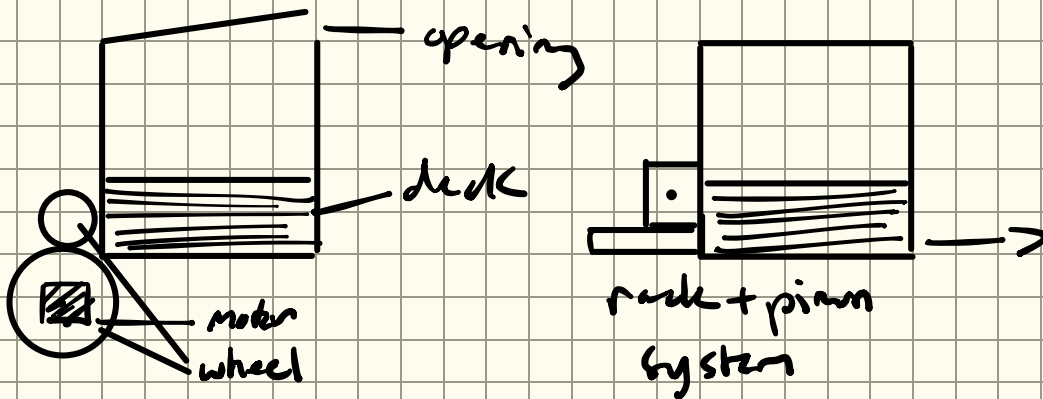
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Rotating Subsystem:

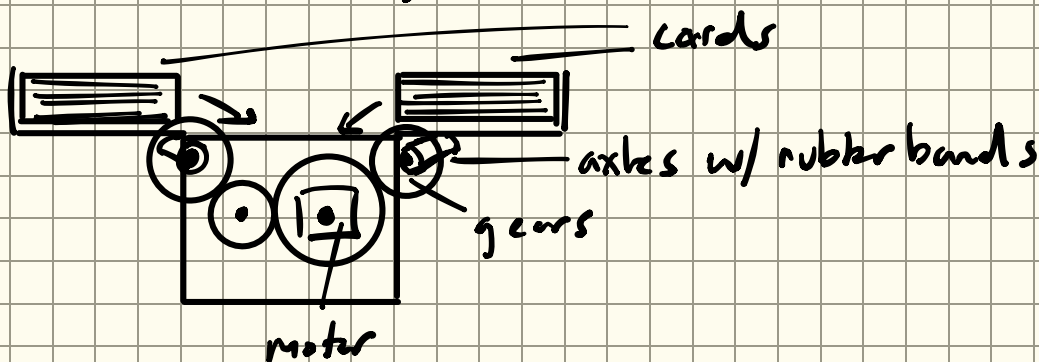


Card deck $\Rightarrow$ heavy $\Rightarrow$ need to be durable
 servo is weak } gear box $\Rightarrow$ increase torque
 $\Rightarrow$ increase stability

Card deck Container:



Shuffler Subsystem:



Designer: Ethan Luong

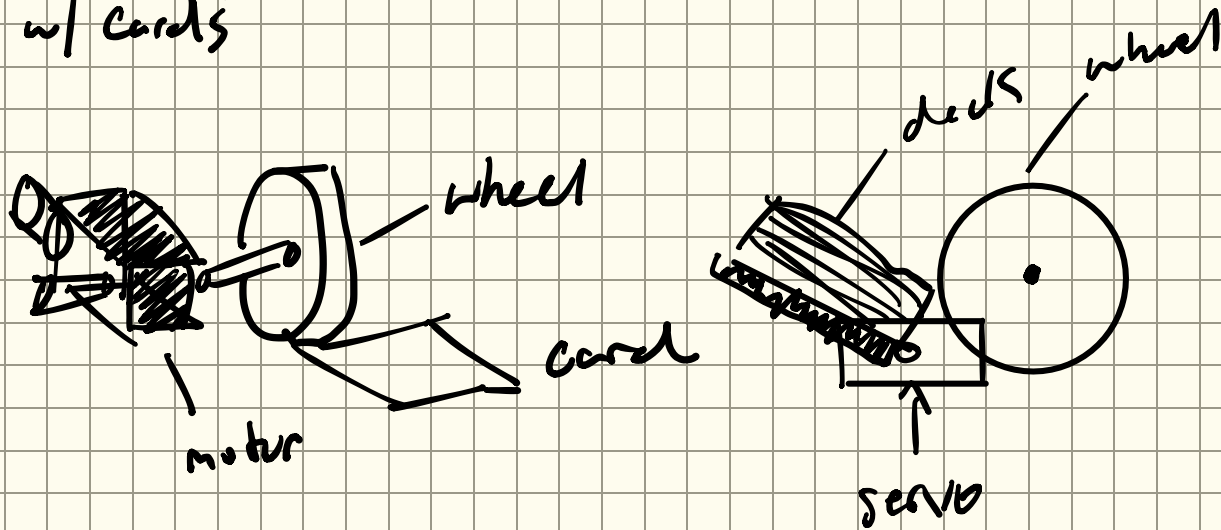
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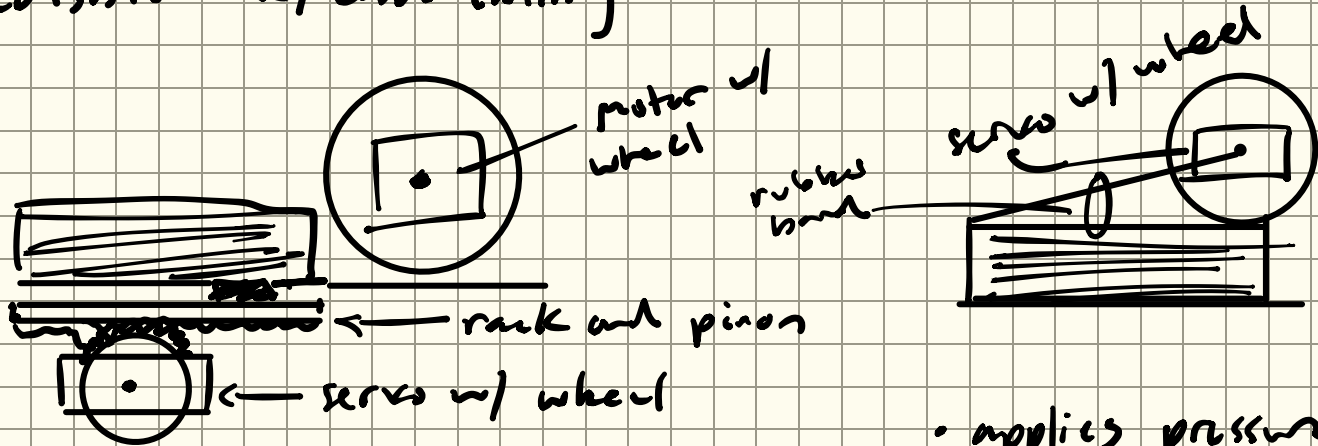
Date: _____

1. Test motors
2. Build test subsystem
3. Test w/ cards

1.



- Uses gravity to let a card into spinning wheel
- inconsistent w/ card falling



- push on card at a time consistent
- needs material w/ a lot of friction
- more complex / less compact

- applies pressure passively
- wheel quick spins out one
- less complex and more compact
- cards can't fall in

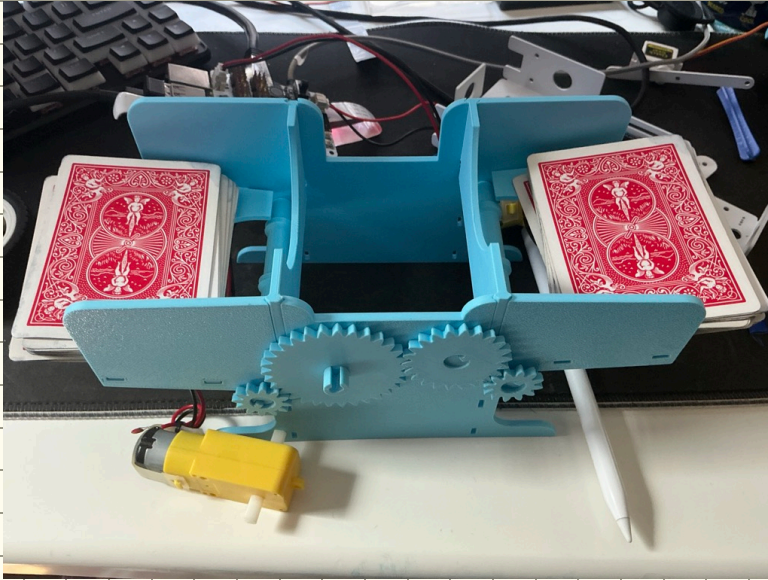
Designer: Ethan Luong

Witness:

Date: 7/12/26

Date:

Question: Is one or two motor a better solution to shuffle the two-halves of a deck?



- Mechanically complex
- Requires one motor
- Need precise tolerance to spin gear train freely
- Need o-rings
- Fixed shuffling speeds

- Electronically complex
- Requires two motors
- Can vary motor speed for random shuffling
- Need o-rings or rubber surfaces
- Won't suffer from too much friction to spin

Verdict: Two motor solⁿ is better for consistency and more room for electronic/software improvements

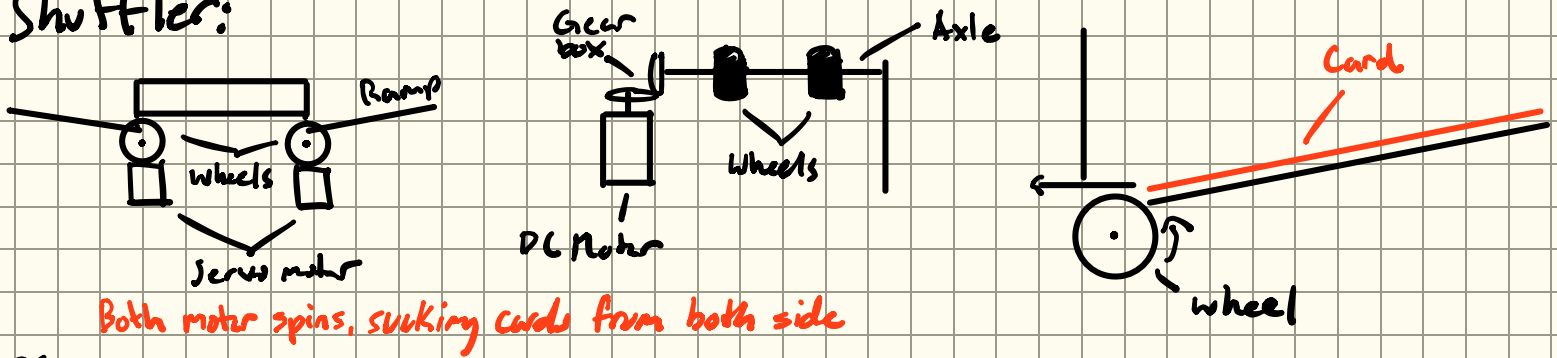
Designer: Ethan Luong

Date: 7/14/26

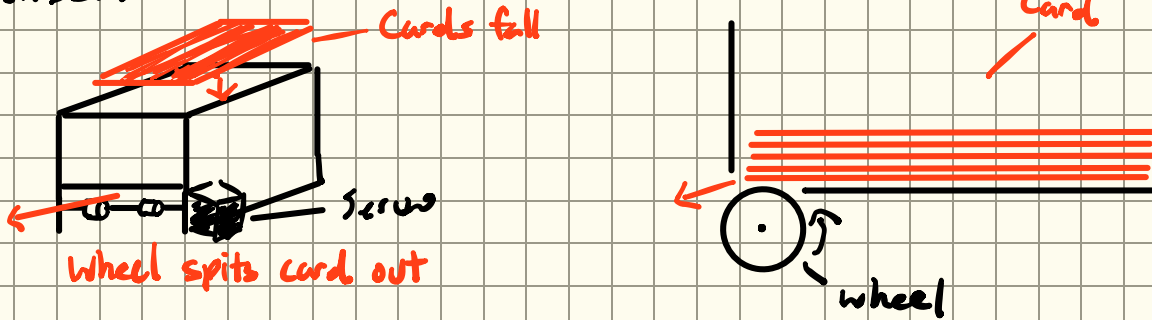
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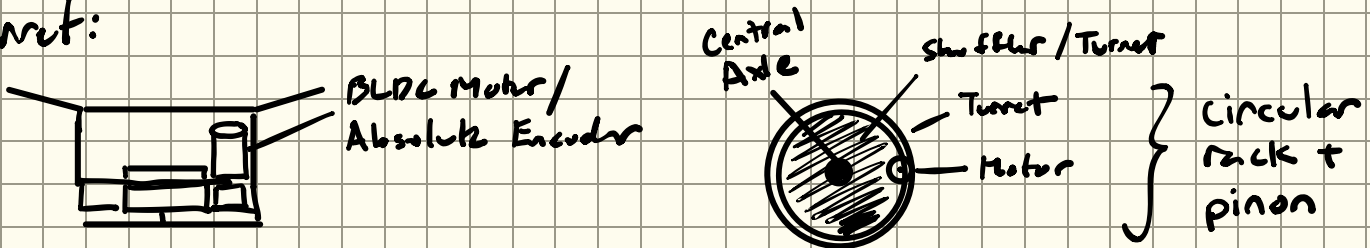
Shuffler:



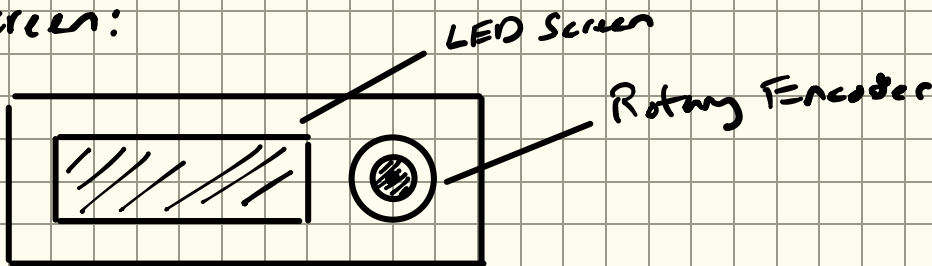
Dispenser:



Turner:



Screen:



Functions:

1. Select Deal Settings
2. Start Shuffle
3. Start Dealing
4. Calibrate Player's positions

Designer: Ethan Luong

Witness: _____

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Date: _____

Electrical Power Budget

6

Battery : 3 18680 3.7V Battery Cells (11.1V nominal / 12.6 V max)

Actuators	Voltage	Max/Stall Current	No load Current
GoBilda Servo	4.8 - 7.4V	2000 - 3000 mA	200 mA
BLDC Motor	7.4 - 16V	4.4A	~0.4 mA

Components	Voltage	Max Current	Power line	Total Curr.
ESP32	5V	500 mA	12V	1 A
1602 LCD	5V	30 mA	6V	7.5 A
ASS600 Mag. Enc.	3.3V	6.5 mA	5V	0.6 A
KY-040 Enc.	5V	< 40 mA	4.4A < motor driver peak current 7.5 A < buck converter peak current 0.6 A < buck converter peak current	
Total	5V	600 mA		

Components	Avg. Current
Servos	0.2 x 3 A
Motor	0.4 A
ESP32+Others	~0.2 A

$$\text{Total Avg Current} = 0.6 + 0.4 + 0.2 = 1.2 \text{ A}$$

Runtime	Min. Capacity	Ideal Capacity (20% Buffer)
0.5h	600 mAh	720 - 800 mAh
1h	1200 mAh	1440 - 1500 mAh
2h	2400 mAh	2880 - 3000 mAh
3h	3600 mAh	4320 - 4500 mAh

Suppose the average runtime of this machine is 1-1.5 hours.
 Then, the target battery capacity is between 2000 - 2100 mAh.
 If we have a 11.1V/12.6V 3000 mAh battery, then max runtime is about 2-2.5 hours, which exceeds the average runtime.

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Date: 7/19/26

Witness: _____

Date: _____

How much torque is needed to spin the chassis?

Components	Weight
Servos	3 x 60g
DC Motor	148g
Battery	3 x ~46g
Converter 10A	18g
Converter 3A	80g
ESP32	32g
Rotary Enc.	~50g
LCD Display	150g
Boards	3 x ~5g
Total:	861g

$$\text{Max Weight} = 1.4 \text{ Kg} = 1400 \text{ g}$$

$$\text{Max Chassis Weight} = 1400 \text{ g} - 861 \text{ g} = 539 \text{ g}$$

$$\text{Base plate (non rotating) weight} = \text{volume (cm}^3) \cdot 1.24 \text{ g/cm}^3 \cdot \text{Infill \%} \cdot \text{Boff} = \pi h (R_o^2 - R_i^2) \cdot 1.24 \cdot I (\text{g})$$

Let say $h = 1 \text{ cm}$, $I = 0.45$, $R_o = 15 \text{ cm}$, and $R_i = 10 \text{ cm}$. Then, the weight is about 70-85 g.

Old Calculation for previously chosen motor! New motor has way more torque so gear reduction

Estimated Weight on Turret = 1 Kg *is not needed!*

$$\begin{aligned} \tau &= I \alpha & I &= \frac{1}{2} M R^2 = \frac{1}{2} \cdot 1 \cdot 0.15^2 = 0.01125 \text{ Kg} \cdot \text{m}^2 \\ &= 0.01125 \cdot 25.12 & \omega &= \text{RPM} \cdot \frac{2\pi}{60}, \text{ suppose we want turret to} \\ &= 0.2826 \text{ N} \cdot \text{m} & &= 6.28 \text{ rad/s} \text{ spin to 60 RPM in } 0.25 \text{ s} \\ \alpha &= \frac{\omega}{t} = \frac{6.28}{0.25} = 25.12 \text{ rad/s}^2 \end{aligned}$$

To account for friction, apply safety factor of 2 to ideal torque.

$$\tau' = 0.5652 \text{ N} \cdot \text{m} \quad \tau_m = \frac{\tau'}{GR} \Rightarrow GR = \text{Gear Ratio} = 19.22$$

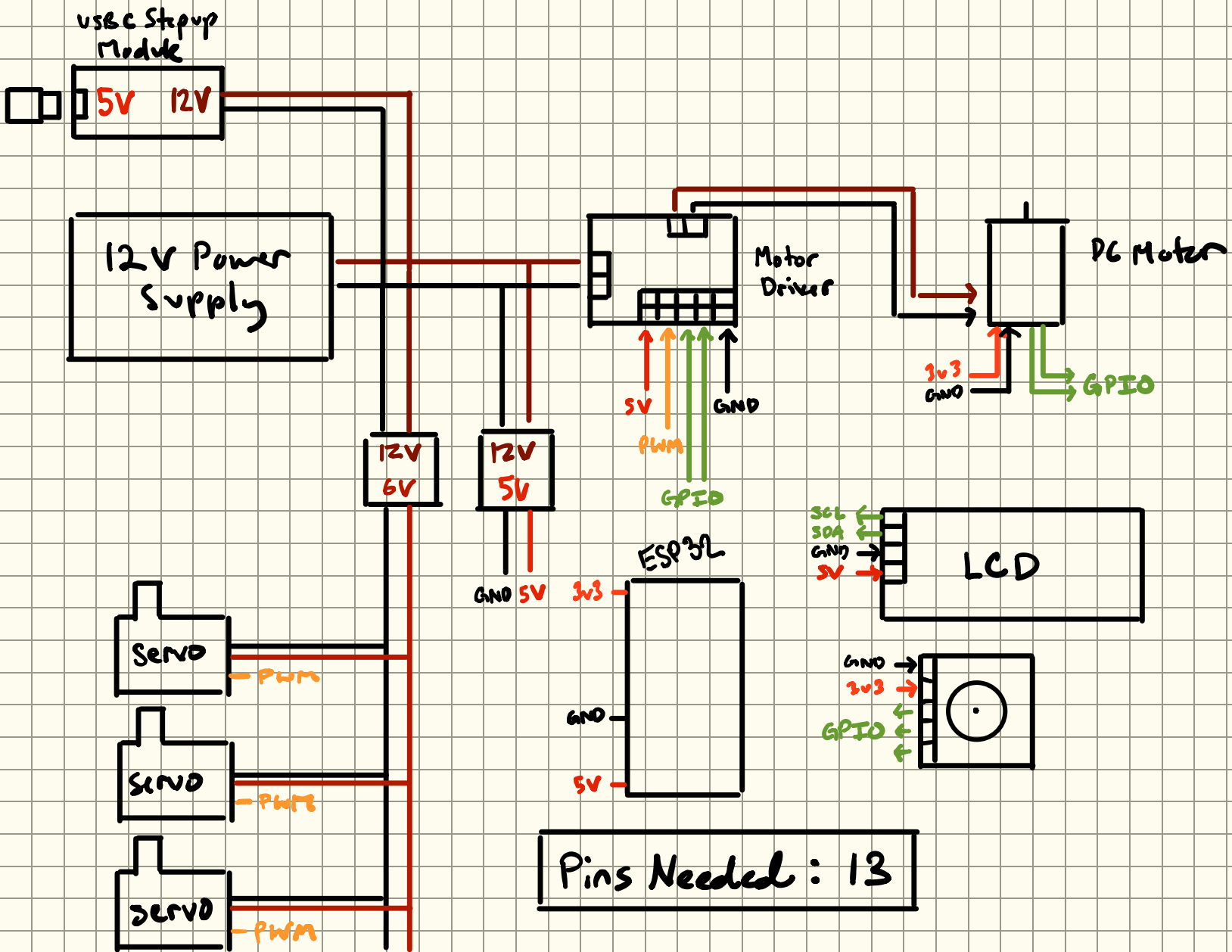
$$\tau_{\text{motor}} \approx 0.0294 \text{ N} \cdot \text{m}$$

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Date: 7/19/26

Witness: _____

Date: _____



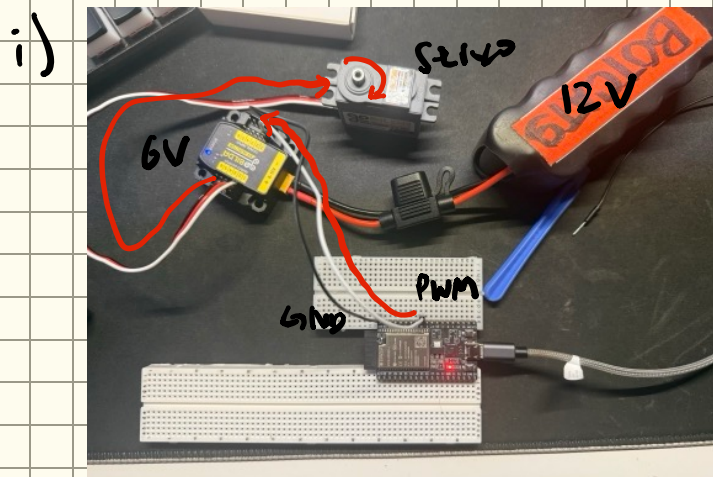
Designer: Ethan Luong

Witness:

Date: 7/20/26

Date:

- Objectives:
- i) Test all usable ESP32 GPIO pin using servo + pwm
 - ii) Verify direction + speed control of DC Motor driver
 - iii) Perform RPM test on motor for ideal shuffling spl



Created a simple circuit and program that send a PWM signal to the servo via a gpio pin. If the servo shaft moves, then that gpio pin is working (then repeat for every usable pin)

In the program, I had to specify a frequency, duty value, and its duty resolution. Here's the settings I used:

f: 50 hz (20ms/20000µs)

Duty Res: 14 bits (0-16383)

PWM Range (PW): 500 - 2500 µs
(0 - 300 degrees)

Duty Range = PW · f · Duty Res

Min: 410 (500 µs, 0 deg)

Mid: 1227 (1500 µs, 150 deg)

Max: 2048 (2500 µs, 300 deg)

Result: PASSED

Notes: Pin 16+17 are not usable gpio pins! Other than that, all usable GPIO pins were able to send a pwm signal.

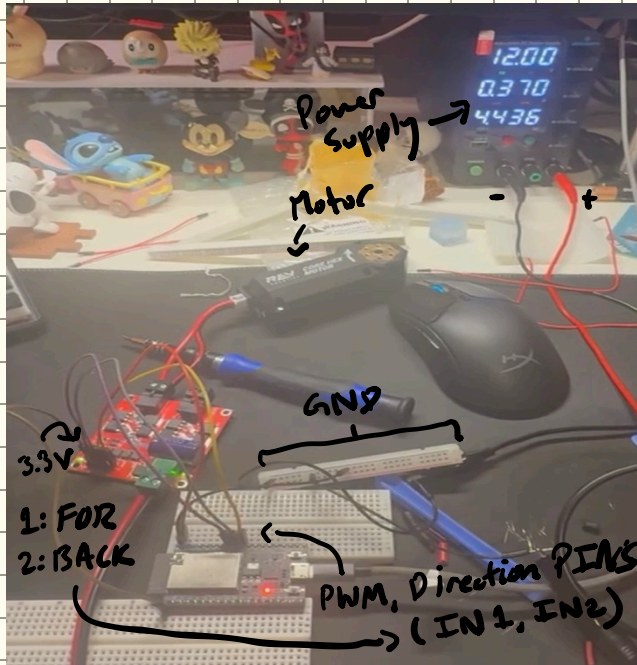
Designer: Ethan Luong

Date: 7/23/26

Witness:

Date:

ii)



Create a circuit + program that simply send a pwm signal for speed not a high/low signal for direction via 3 gpio pins. Once again, I need to send a duty value for pwm. So, here what I did:

Duty Rms = 10 bit (0-1023) $\Rightarrow$ Duty Value (0-1023)

Forward: set IN1 high, set IN2 low, set duty value

Backward: set IN2 high, set IN1 low, set duty value

Stop: set IN1 and IN2 low, set duty value.

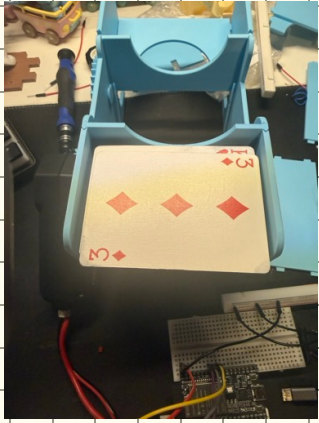
Result: The motor is able to rotate forward and backwards! I did find out that the motor is super loud, so it is likely that I need to run at low speeds to reduce noise. The amount of torque it has is a lot and only drew < 1 A from me grabbing the tire on the shaft tightly (Max 4.4A). So, I can direct drive or even gear overdrive slightly to run at the perfect RPM for the turret!

Designer: Ethan Lunny

Date: 7/23/26

Witness: _____

Date: _____



What is the ideal RPM for shuffling? Gear ratio?

Test: Put a motor with a tire on its shaft under a platform with a full deck of cards. Record the time it takes to dispense a full deck at a running rpm.

Then, half the time to get the ideal RPM that translate to an ideal shuffle time. Lets say ideal shuffle time is 2-4 seconds

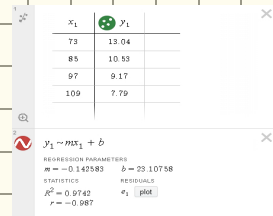
$$\text{RPM} = \text{Max RPM} \times \frac{\text{Duty}}{\text{Max Duty}} \quad \left(\begin{array}{l} \text{Max RPM: 125 RPM} \\ \text{Max Duty: 1023} \end{array} \right)$$

$$600 \text{ duty} (\sim 73 \text{ rpm}) \Rightarrow 26.08 \text{ sec} \stackrel{\div 2}{\Rightarrow} 13.04 \text{ sec}$$

$$700 \text{ duty} (\sim 85 \text{ rpm}) \Rightarrow 21.05 \text{ sec} \stackrel{\div 2}{\Rightarrow} 10.53 \text{ sec}$$

$$800 \text{ duty} (\sim 97 \text{ rpm}) \Rightarrow 18.33 \text{ sec} \stackrel{\div 2}{\Rightarrow} 9.17 \text{ sec}$$

$$900 \text{ duty} (\sim 109 \text{ rpm}) \Rightarrow 15.58 \text{ sec} \stackrel{\div 2}{\Rightarrow} 7.79 \text{ sec} \quad \left. \vphantom{\begin{array}{l} 600 \\ 700 \\ 800 \\ 900 \end{array}} \right\} \sim -0.143 \frac{\text{s}}{\text{RPM}}$$



$$\text{Max Servo RPM (@ 6V): 115 RPM} \Rightarrow \sim 6.71 \text{ sec}$$

Even at the max speed in continuous mode, it is slightly slow assuming the exact setup. That means a gear overdrive is necessary to achieve 2-4 sec shuffle time.

$$2 \text{ sec} \Rightarrow 148 \text{ RPM}, 115:148 \Rightarrow 1:\sim 1.29$$

$$4 \text{ sec} \Rightarrow 134 \text{ RPM}, 115:134 \Rightarrow 1:\sim 1.17$$

$$\text{Ideal RPM: } \boxed{134 - 148 \text{ RPM}}, \text{ Ideal GR: } 1:1.17 - 1:1.29$$

Designer: Ethan Luong

Date: 7/23/26

Witness: _____

Date: _____

V1 Setup:

Wheel Diameter: 32 mm

Gear Ratio: 1:3

Results:

Heights: 50 mm, 85 mm

Distance: 5 inches, 7 inches

V2 Setup:

Wheel Diameter: 48 mm

Gear Ratio: 1:6.5

Results:

Heights: 65 mm

Distance: 16 inches

V1 Problems:

- Gap too boxy for card to smoothly dispense out
- gear overdrive needs to be larger, card not launched far
- shaft tolerance too small, shaft move around too much
- Fixed gate doesn't allow for experimental gate gaps

V2 Improvements:

- Adjustable gate with fillet gap
- 1:6.5 overdrive
- 48 mm gear wheel for more torque and grip
- Stable shaft with more tolerance

V2 Problems:

- Not enough contact with wheel

Designer: Ethan LuongDate: 7/27/26

Witness: _____

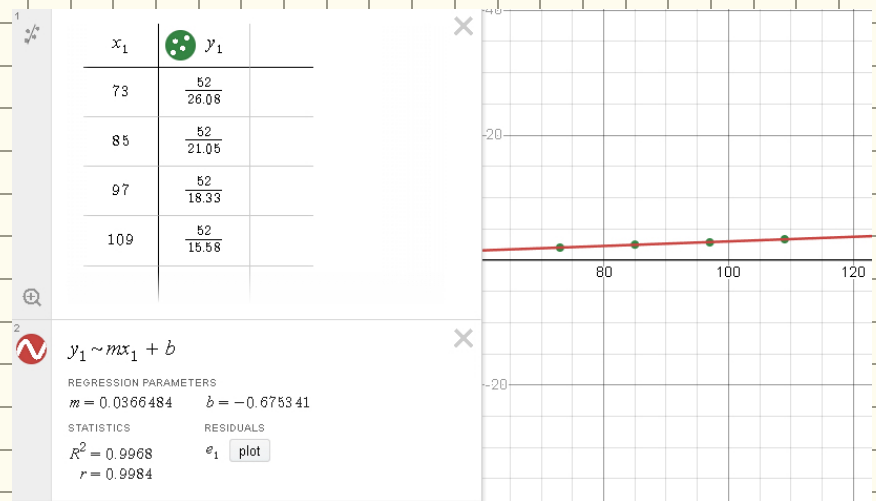
Date: _____

My calc on page 11 was made on two wrong assumptions:

1. Time and RPM is not linear, it is a hyperbola relationship ($t = \frac{a}{\text{RPM}}$)
2. I halved the time before doing a linear regression which gave me a rate half of which it actually is

Corrections:

$$\begin{aligned} \text{Rate} &= \frac{\text{card}}{t}, t = \frac{a}{\text{rpm}} \\ &= \frac{\text{card}}{a/\text{rpm}} \\ &= \frac{\text{card} \cdot \text{rpm}}{a} \\ &= \left(\frac{\text{card}}{a}\right) \cdot \text{rpm} \end{aligned}$$



$$\text{Rate} = k \cdot \text{rpm (linear!)} \Rightarrow f(\text{rpm}) = k \cdot \text{rpm} + b = \text{rate}$$

$$\text{Rate}_{4\text{sec}} = \frac{26 \text{ cards}}{4 \text{ secs}} = 6.5 \text{ c/s}, f^{-1}(6.5) \approx 196 \text{ rpm}$$

$$\text{Rate}_{2\text{sec}} = \frac{26 \text{ cards}}{2 \text{ secs}} = 13 \text{ c/s}, f^{-1}(13) \approx 373 \text{ rpm}$$

$$\text{GR}_{4\text{sec}} = \frac{196}{115} = 1.7$$

$$\text{GR}_{2\text{sec}} = \frac{373}{115} = 3.25$$

Ideal RPM: 196 - 373 rpm

Ideal GR: 1:1.7 - 1:3.25

Designer: Ethan Luong

Witness: _____

Date: 7/30/26

Date: _____

Shuffler Test

148

V1 Setup:

Wheel Diameter: 32 mm

Gear Ratio: 1:1.5

Results:

Time: ~10 seconds

V1 Problems:

- Shaft tolerance too loose
- Incorrect gear ratio based on initial calculations
- Wheel tread design too slanted, causes sideways motion
- Wheel material has too low coefficient of friction to pull bottom card out

V2 Setup:

Wheel Diameter: 14 mm

Gear Ratio: 1:1.5

Results:

N/A, too inconsistent so could not get a solid shuffle time

V2 Improvements:

- Moved servos to the top
- Used custom adhesive foam wheel
- Changed gear ratio to 1:3
- Deck angle from $15^\circ \Rightarrow 20^\circ$

V2 Problems:

- Foam strip peels off due to contact with wall and gate
- Too little card contact

Designer:

Ethan Luong

Date:

8/4/26

Witness:

Date:

V3 Setup:

Wheel Diameter: 19 mm

Gear Ratio: 1:3

V3 Improvement:

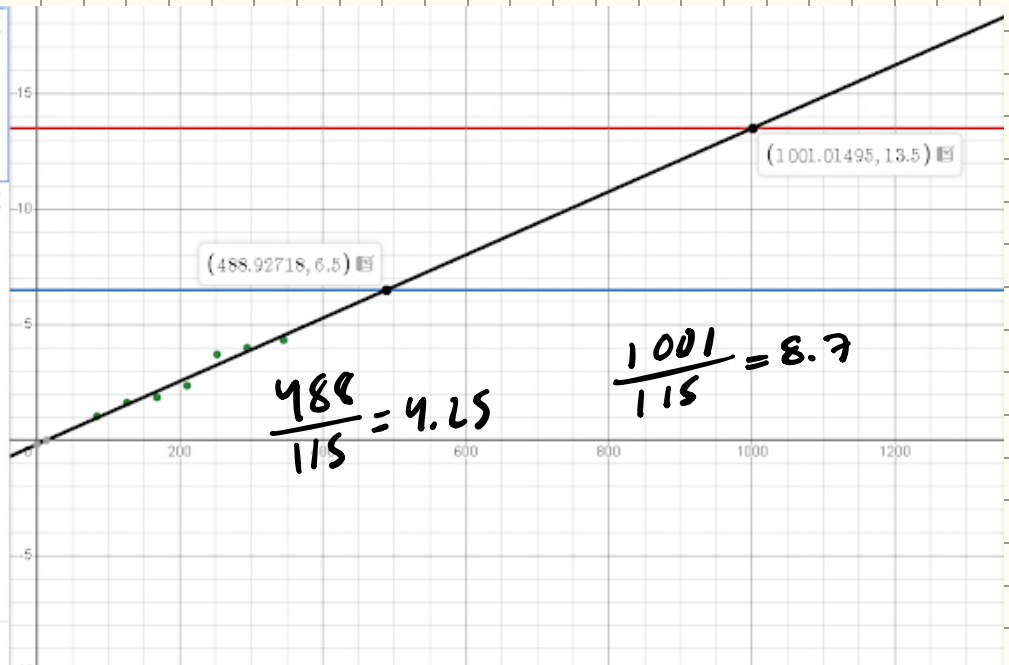
- Thinner gate for more card contact
- Changed frame strip orientation to be perpendicular to cards

Speed	Time
87 rpm	25 sec
126 rpm	16 sec
168 rpm	14 sec
210 rpm	11 sec
252 rpm	7 sec
294 rpm	6.5 sec
345 rpm	6 sec

V3 Problems:

- Still slow shuffle time
- non-rigid gate
- side plates has weak structural points

New Ideal Gear ratio: 1:4.25 - 1:8.7



Designer: Ethan Luong

Date: 8/17/26

Witness:

Date:

V4 Set up:

Wheel Diameter: 19 mm

Gear Ratio: 1:9

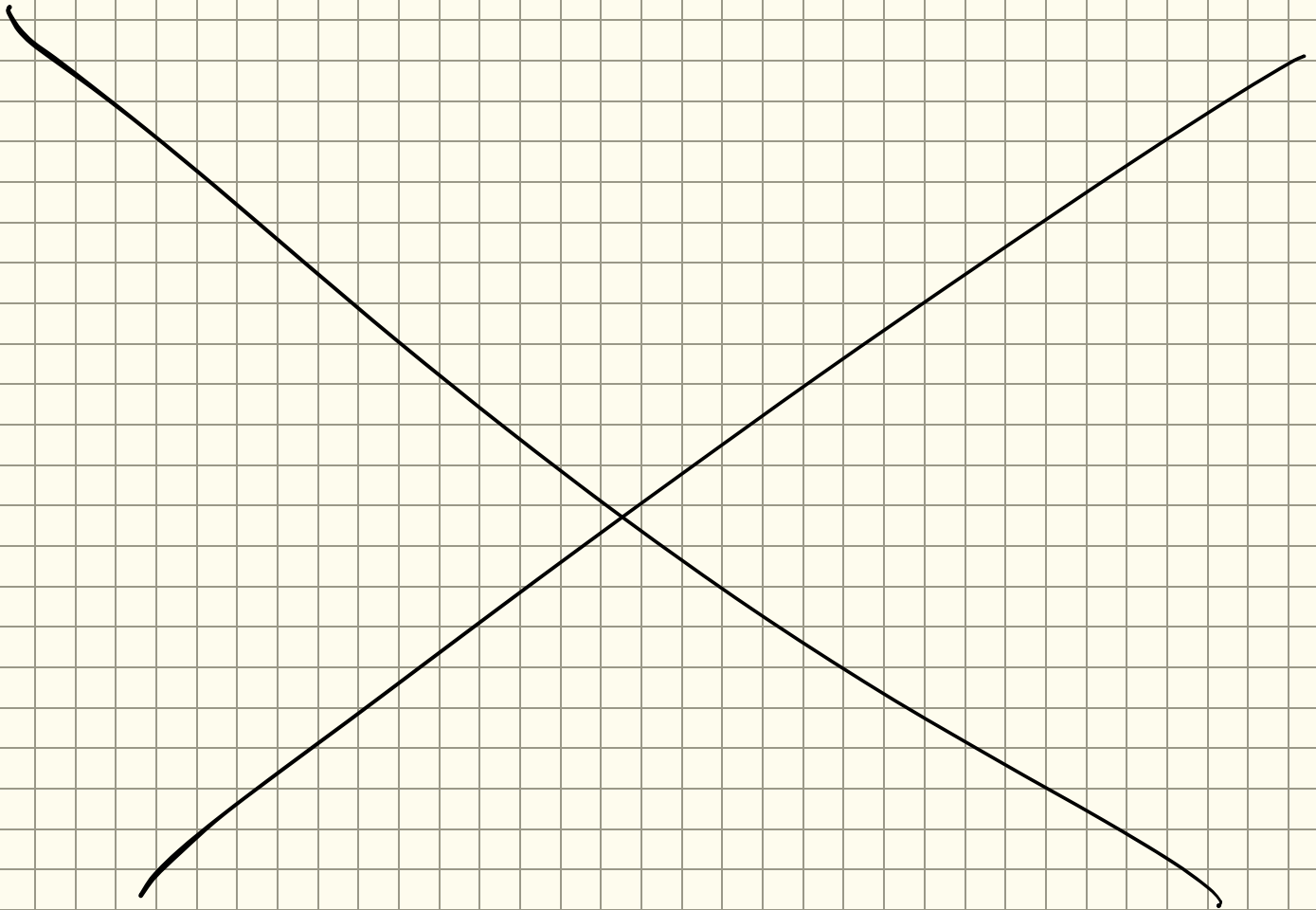
Result: 2-4 seconds

V4 Improvement:

- Rigid Gate and inner wall
- Redesigned side plates for compound gear train

V4 Problems:

- Cards get jammed + collides
- Deck space height too low



Designer:

Ethan Luong

Date:

8/18/26

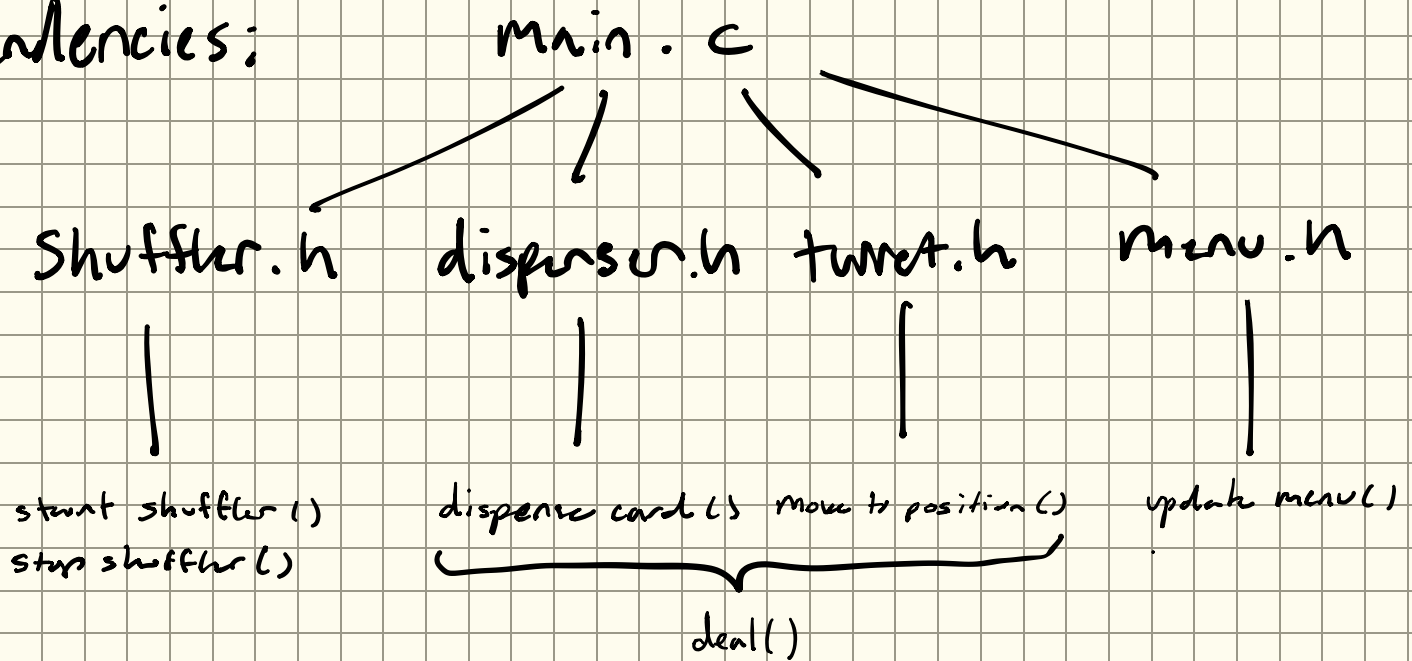
Witness:

Date:

Subsystems: shuffler, dispenser, turret, menu

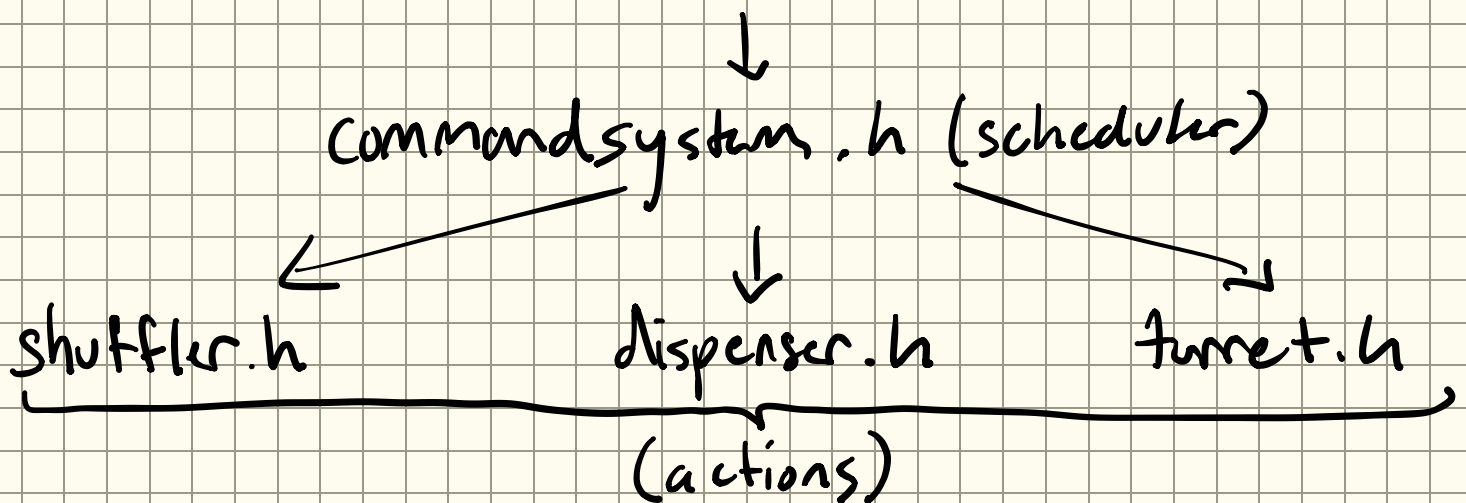
Actions: start shuffler(), stop shuffler(),
dispense card(), move to position(),
update menu()

Dependencies:



I/O flow:

menu.h (inputs)



Designer:

Ethan Luvay

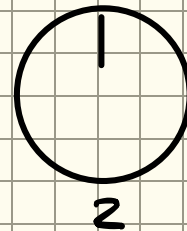
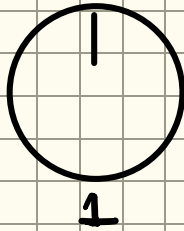
Date: 9/6/26

Witness:

Date:

Menu Layout:

Mode: _____	[111]	
Player	Cards	Extra
4	2	Ø



Menu Pseudo Code:

- if button 2 pressed: edits players $\rightarrow$ cards $\rightarrow$ extra
- if button 2 rotated: increases current value
- if button 1 rotated: cycle between shuffle $\rightarrow$ deal $\rightarrow$ both
- if button 1 pressed: perform said action

Designer:

Ethan Luong

Date:

9/6/26

Witness:

Date:

To do:

1. Improve shuffler jams

2. Redesign dispenser

- i) Switch to equivalent compound gear train + smaller wheel
- ii) Add a small-angled ramp to deal with bowed cards
- iii) Add opening in the back for retrieving deck

3) Construct power system

- i) Build the 12.6 V battery w/ BMS module
- ii) Build SPDT switch between charging pole (off) and main pole (on)
- iii) Design low-voltage shutdown sequence

4) Redesign turner

i) Switch motor to continuous servo motor $\frac{1}{2}$ absolute magnetic encoder

ii) Redesign turner code for a brushed servo motor

5) Add sensors

- i) IR sensors for directional dealing
- ii) distance sensor for shuffling timing (optional)

6) Design and assemble PCB

i) holds ESP32, shutdown circuit, power lines

Designer: Ethan Luong

Date: 9/9/26

Witness: _____

Date: _____

Lard Dispenser Gear Ratio & Gear Modules

20

Gear Ratio:

$$D_o = 48 \text{ mm} \quad GR_o = \frac{1}{6.5}$$

$$D_f = 32 \text{ mm} \quad GR_f = ?$$

$$V = WR = \left(\frac{W_{\text{motor}}}{GR} \right) \left(\frac{D}{2} \right)$$

$$V_o = \left(\frac{W_{\text{motor}}}{GR_o} \right) \left(\frac{D_o}{2} \right)$$

$$V_f = \left(\frac{W_{\text{motor}}}{GR_f} \right) \left(\frac{D_f}{2} \right)$$

$$V_o = V_f$$

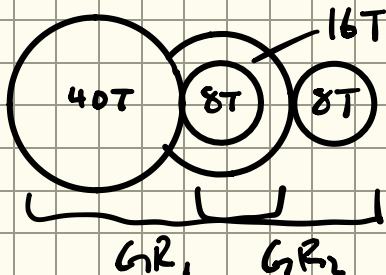
$$\left(\frac{W_{\text{motor}}}{GR_o} \right) \left(\frac{D_o}{2} \right) = \left(\frac{W_{\text{motor}}}{GR_f} \right) \left(\frac{D_f}{2} \right)$$

$$\left(\frac{D_o}{GR_o} \right) = \left(\frac{D_f}{GR_f} \right)$$

$$\frac{48}{\frac{1}{6.5}} = \frac{32}{GR_f}$$

$$GR_f = \frac{1}{9.75}$$

Gear Modules:



$$GR_1: 1:5 \quad d = \text{center-to-center distance}$$

$$GR_2: 1:2 \quad m = \text{gear module}$$

$$2 \cdot d = (T_{\text{out}} + T_{\text{in}}) \cdot m$$

GR_1

GR_2

$$d_1 = 31 \text{ mm}, T_{\text{out}} = 8, T_{\text{in}} = 40$$

$$m_1 = \frac{2 \cdot d_1}{T_{\text{out}} + T_{\text{in}}} = \frac{2 \cdot 31}{48} = \frac{62}{48} = \frac{31}{24}$$

$$d_2 = 16 \text{ mm}, T_{\text{out}} = 8, T_{\text{in}} = 16$$

$$m_2 = \frac{2 \cdot d_2}{T_{\text{out}} + T_{\text{in}}} = \frac{2 \cdot 16}{24} = \frac{32}{24} = \frac{4}{3}$$

Designer: Ethan Luong

Date: 9/10/26

Witness: _____

Date: _____